

CHAPTER 10: THE SHAPES OF MOLECULES

Lewis Symbols (Chapter 9.1)

- For a main group element, the group number (1A – 8A; 1 – 8) is the valence number of electrons in the atom.
- This is how many dots are needed in the electron dot symbol.
- Add dots 1 at a time on each side of the element symbol.
- Pair the dots up once all four sides have 1 dot until no valence electrons are left.

Oxygen – 6 valence electrons



Octet Rule

- Atoms gain, lose or share electrons when bonding to fill the outer valence shell with 8 electrons.
- Hydrogen and Lithium only hold 2 electrons in their outer valence shell.
- Applies to nearly all period 2 elements and many others.

Lewis Structures

- A structural formula which consists of electron-dot symbols.
- Bonding pairs are indicated with a line
- Lone pairs are indicated as dots



Writing Lewis Structures

- Place atoms relative to each other
 - Atom with the lower group number is placed in the center. (CCl_4)
 - Needs more electrons to attain an octet, usually has lower electronegativity.
 - If atoms are in the same group number, the atom with the higher period number goes in the center. (SO_3)
 - Hydrogen is never a central atom since it only forms 1 bond.

Writing Lewis Structures

- Determine the total number of electrons the molecule needs (N):
 - $N = 8 \times (\text{\# of non-hydrogen atoms}) + 2 \times (\text{\# of hydrogen atoms})$
 - Ex.
 - CCl_4

Writing Lewis Structures

- Determine the total number of valence electrons available (A)
 - CCl_4
 - C – 4 valence electrons
 - Cl – 7 valence electrons
 - $[1 \times \text{C}(4e^-)] + [4 \times \text{Cl}(7e^-)] = 32 \text{ valence } e^-$
 - For a polyatomic ion you add one electron for each negative charge or subtract one electron for each positive charge
 - BF_4^-
 - NH_4^+

Writing Lewis Structures

- Determine the number of Shared electrons in the molecule (S):
 - $S = N - A$
 - Ex:
 - CCl_4
- Determine the number of bonding pairs:
 - $\text{bp} = \frac{S}{2}$
- Determine the number of lone pairs of electrons to complete octets of non-hydrogen atoms:
 - $\text{lp} = \frac{A}{2} - \frac{S}{2} = \frac{A}{2} - \text{bp}$

Writing Lewis Structures

- Draw a single line for each bonding pair from the central atom to an atom surrounding the central atom.
- Distribute the lone pairs of electrons so each atom ends up with 8 electrons.
 - Place lone pairs on surrounding atoms (more electronegative atoms first).
 - If electrons remain after the surrounding octets are full place the remaining electrons on the central atom.
- Ex.
 - BF_4^-
 - NH_4^+

Writing Lewis Structures

- Write the Lewis Structure for:

- a) OF_2
- b) CH_2Br_2
- c) AlCl_4^-

- Write the Lewis Structure for:

- a) NH_3O (hydroxylamine)
- b) $\text{C}_2\text{H}_6\text{O}$ (dimethyl ether)

Writing Lewis Structures

- If a central atom does not form an octet then one or more multiple bonds needs to be formed.
- Ex.
 - O_2
 - CO
 - N_2H_2

Resonance Structures

- Resonance is the delocalized sharing of electrons.
 - The electron pair is spread out over a few atoms, not just shared between 2.
- Resonance structure – one of two or more Lewis Structures for a species that cannot be depicted by a single structure.
 - These structures differ in the position of bonding and lone electron pairs only!!
 - Ex.
 - O_3
 - NO_2
 - The molecule is a resonance hybrid (average) of the resonance structures.

Bond Orders

- A bond order is the number of electron pairs a specific number of atoms shares.
- In resonance structures partial bonding is observed.
- You can calculate the bond order with the calculation:
 - Bond order = $\frac{\text{Number of electron pairs}}{\text{Number of bonded atoms}}$
 - Ex.



Formal Charge

- The hypothetical charge an atom would have if bonding electrons were equally shared.
 - Formal charge = # of valence e - (# unshared valence e- + $\frac{1}{2}$ # of shared valence e-)
 - $FC = V - (LP \text{ electrons} + \frac{BP \text{ electrons}}{2})$
- Ex.
 - CO_3^{2-}

Which Resonance Structure is Most Important?

- The Major resonance structure will have:
 - Smaller formal Charge
 - The same non-zero formal charges are not on adjacent atoms
 - The more negative formal charge is on the more electronegative atom.
- Ex.
 - NCO^-

Exceptions to the Octet Rule

- Central Atom is an Electron-Deficient Atom
 - Molecules in which beryllium or boron is the central atom.
 - Fewer than 8 valence electrons
 - Ex.
 - BeCl_2
 - BF_3
 - React with other compounds to obtain an octet.
 - Ex.
 - $\text{BF}_{3(g)} + \text{NH}_{3(g)}$

Exceptions to the Octet Rule

- Central atom has an Odd Number of Electrons
- Molecules contain one unpaired electron
 - Is paramagnetic and considered a free radical
 - Extremely reactive with hydrogen in biomolecules causing gene disruption
 - Ex.
 - $\text{NO}_{2(g)} + \text{NO}_{2(g)}$

Exceptions to the Octet Rule

- Central Atom can have an Expanded Valence Shell
 - Central Atom must be large and have empty orbitals
 - Occur in non-metals of Period 3 or higher!
 - Central Atom Bonds to more than 4 atoms.
 - Ex.
 - PCl_5
 - SO_4^{2-}
- Central Atom Bonds to fewer than 4 atoms
 - Ex.
 - SF_4